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CSI-Feedback-Driven Link Adaptation in 5G NR Downlink MIMO: A Link- Level Simulation Study

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I hereby declare that, except where specifically indicated, the work submitted herein is my own original work.

Signed Kaiyuan Bao 包铠源 date 30 May 2026

Technical Abstract

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Background and Motivation

Methodology

Key Results and Interpretation

Conclusion

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List of Abbreviations

Abbreviation	Definition
5G	Fifth Generation
BLER	Block Error Rate
CDL	Clustered Delay Line
CP	Cyclic Prefix
CQI	Channel Quality Indicator
CSI	Channel State Information
CSI-RS	CSI Reference Signal
DL	Downlink
DMRS	Demodulation Reference Signal
FR	Frequency Range
gNB	Next Generation Node B
HARQ	Hybrid Automatic Repeat Request
LDPC	Low-Density Parity-Check
LOS	Line-of-Sight
MCS	Modulation and Coding Scheme
MIMO	Multiple-Input Multiple-Output
NLOS	Non-Line-of-Sight
NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing
PDCCH	Physical Downlink Control Channel
PDSCH	Physical Downlink Shared Channel
PMI	Precoding Matrix Indicator
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RB	Resource Block
RE	Resource Element
RI	Rank Indicator
SCS	Subcarrier Spacing
SINR	Signal-to-Interference-and-Noise Ratio
SISO	Single-Input Single-Output
SNR	Signal-to-Noise Ratio
TDL	Tapped Delay Line
UE	User Equipment
UL	Uplink

1 Introduction

2 Background and Methodology

2.1 5G NR Downlink Physical Layer Overview

2.2 MIMO Transmission and CSI Feedback

2.3 CDL Channel Model and Doppler Shift

2.4 MATLAB Simulation Framework

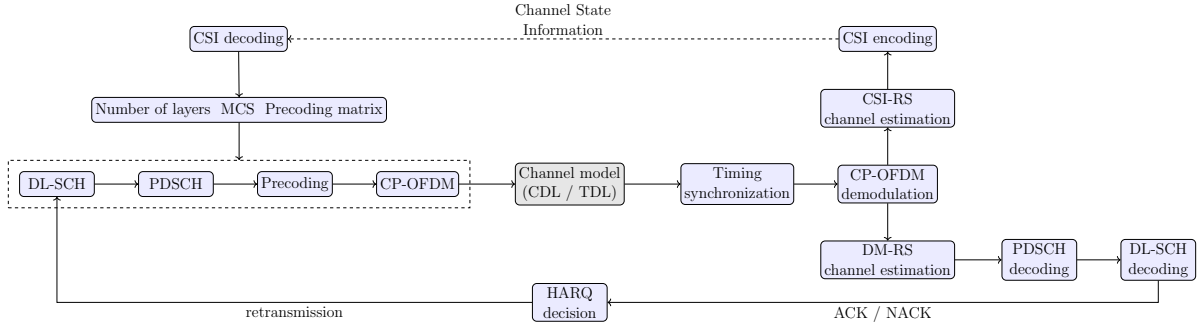


Figure 1: Overview of the CSI-based downlink transmission and feedback framework, adapted from [1]

2.5 Baseline Configuration and Experimental Design

Table 2: System and channel configuration used in the simulation

Parameter	Value
Carrier frequency	FR1, 3.5 GHz
Subcarrier spacing	30 kHz
Channel bandwidth	106 RBs
Antenna configuration	4 Tx, 4 Rx
Maximum number of layers	4
Channel model	CDL-C
Mobility scenarios	Low (10 Hz), High (120 km/h)
Simulation type	Downlink link-level

Table 3: CSI feedback configuration

Parameter	Baseline setting	Investigated values
CSI report periodicity	10 slots	2–40 slots
CQI reporting mode	Wideband	Wideband / Subband
PMI reporting mode	Wideband	Wideband / Subband
Subband size	16 RBs	4–32 RBs
RI configuration	Adaptive	Fixed / Adaptive
Codebook type	Type I	Type I
Codebook mode	Mode 1	Mode 1 / Mode 2

3 Baseline CSI-Feedback Downlink Behaviour

3.1 Baseline CDL-C 4x4 Throughput Behaviour

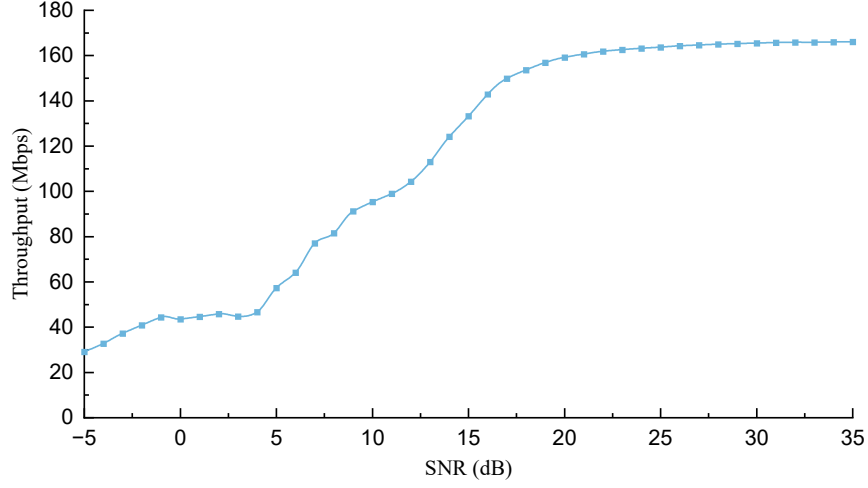


Figure 2

3.2 CSI-driven Link Adaptation Statistics

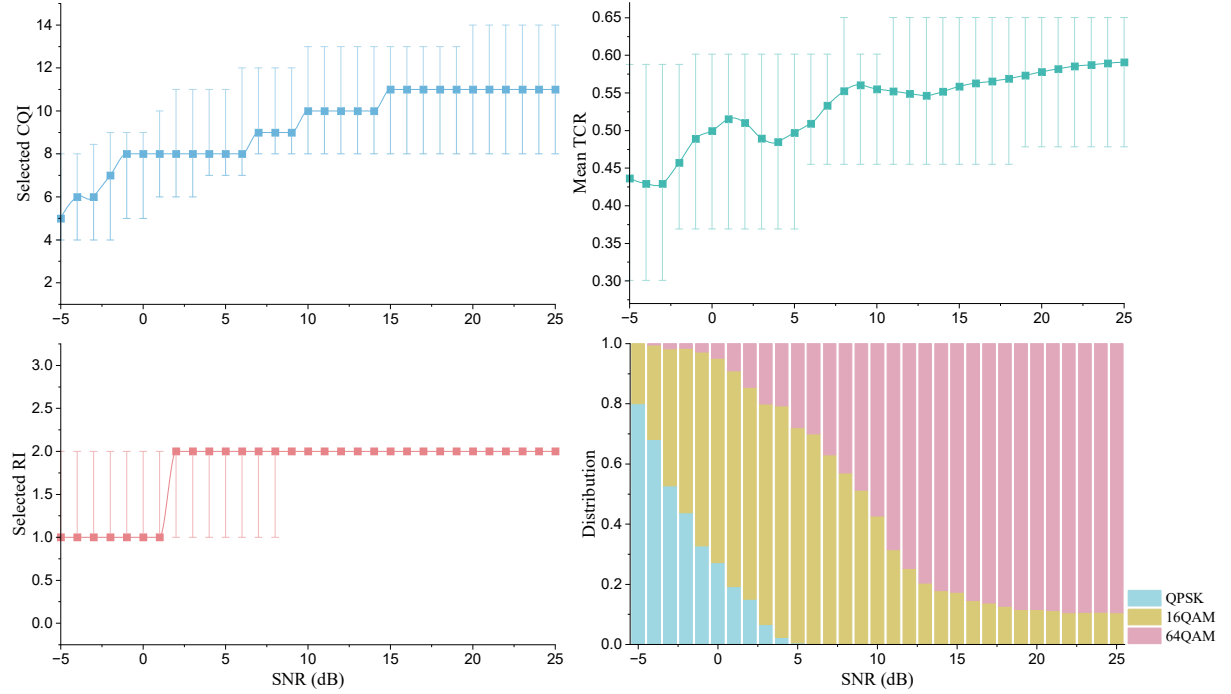


Figure 3

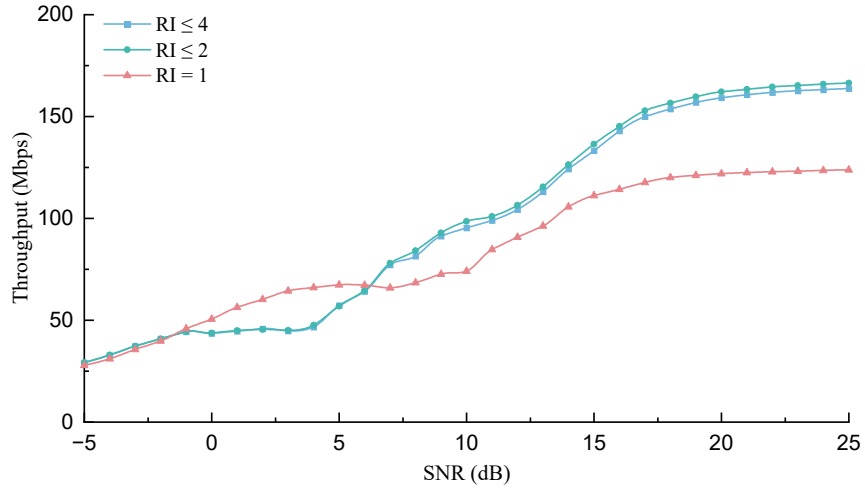


Figure 4: Throughput performance under different maximum layer configurations

3.3 Fixed-Rank Controls

3.4 Summary of Baseline Findings

4 Impact of Mobility and CSI Reporting Periodicity

4.1 Doppler Shift under Default CSI Reporting Period

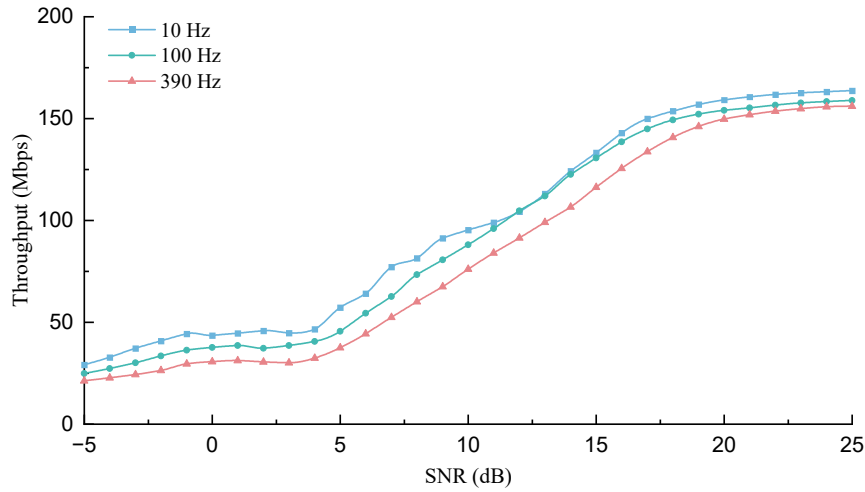


Figure 5

4.2	Reporting Periodicity at Low, Medium and High mobility
4.3	Joint Interpretation: CSI Freshness
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5.1	Antenna Configuration Sweep
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6.1	Interpretation of Results
6.2	Limitations
6.3	Future Work
7	References
 [1] MathWorks, “NR PDSCH Throughput Using Channel State Information Feedback,” 5G Toolbox Documentation. [Online]. Available: https://www.mathworks.com/help/5g/ug/nr-pdsch-throughput-using-csi-feedback.html	
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